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Cluster interpretation document

Objective

The objective of this activity is to apply machine learning techniques to identify hidden patterns in student performance data and segment students into meaningful groups. These segments help mentors understand student risk levels, performance trends, and career readiness beyond rule-based evaluation.

Input Dataset

- **students_scored.csv**
- Generated in Activity 2
- Contains:
 - APS, WWS, PTMS, CRS
 - Student Readiness Index (SRI)
 - Rule-based Risk Category

Machine Learning Approach

Algorithm Used

- **K-Means Clustering (Unsupervised Learning)**

Reason for Selection

- No labeled target variable
- Effective for grouping students based on similarity
- Easy to interpret and mentor-friendly

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Data Preprocessing

Feature Selection

The following features were used for clustering:

- Academic Performance Score (APS)
- Wellness & Wellbeing Score (WWS)
- Productivity & Time Management Score (PTMS)
- Career Readiness Score (CRS)
- Student Readiness Index (SRI)

Normalization

- **StandardScaler** was applied
- Ensures all features contribute equally
- Required for distance-based algorithms like K-Means

Clustering Configuration

Parameter	Value
Algorithm	K-Means
Number of Clusters (k)	4
Initialization	Random
Distance Metric	Euclidean

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Cluster Interpretation

Cluster 0 – At-Risk Students

Characteristics:

- Low APS and SRI
- Poor wellness and productivity
- High stress and distraction levels

Interpretation:

Students in this cluster require immediate academic and wellbeing intervention. They are highly vulnerable to burnout and academic failure.

Cluster 1 – Transitional Students

Characteristics:

- Moderate academic performance
- Inconsistent productivity and wellness
- Mixed SRI categories (Yellow / Blue)

Interpretation:

These students show potential but need structured mentoring to stabilize performance.

Cluster 2 – High Performers

Characteristics:

- High APS and SRI
- Strong productivity and career readiness
- Low stress and high engagement

Interpretation:

Self-driven students suitable for advanced mentoring and leadership development.

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Cluster 3 – Career-Confused Students

Characteristics:

- Strong academics
- Lower CRS compared to APS
- Unclear career goals

Interpretation:

Academically capable students who need focused career guidance rather than academic support.

Comparison with Rule-Based SRI

Observations

- Strong alignment between ML clusters and rule-based SRI categories
- ML revealed additional insights such as career-confused students
- Validates the effectiveness of the rule-based scoring system

Visualization

- Scatter plot of **APS vs SRI**
- Colored by cluster labels
- Clearly shows separation of student groups

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